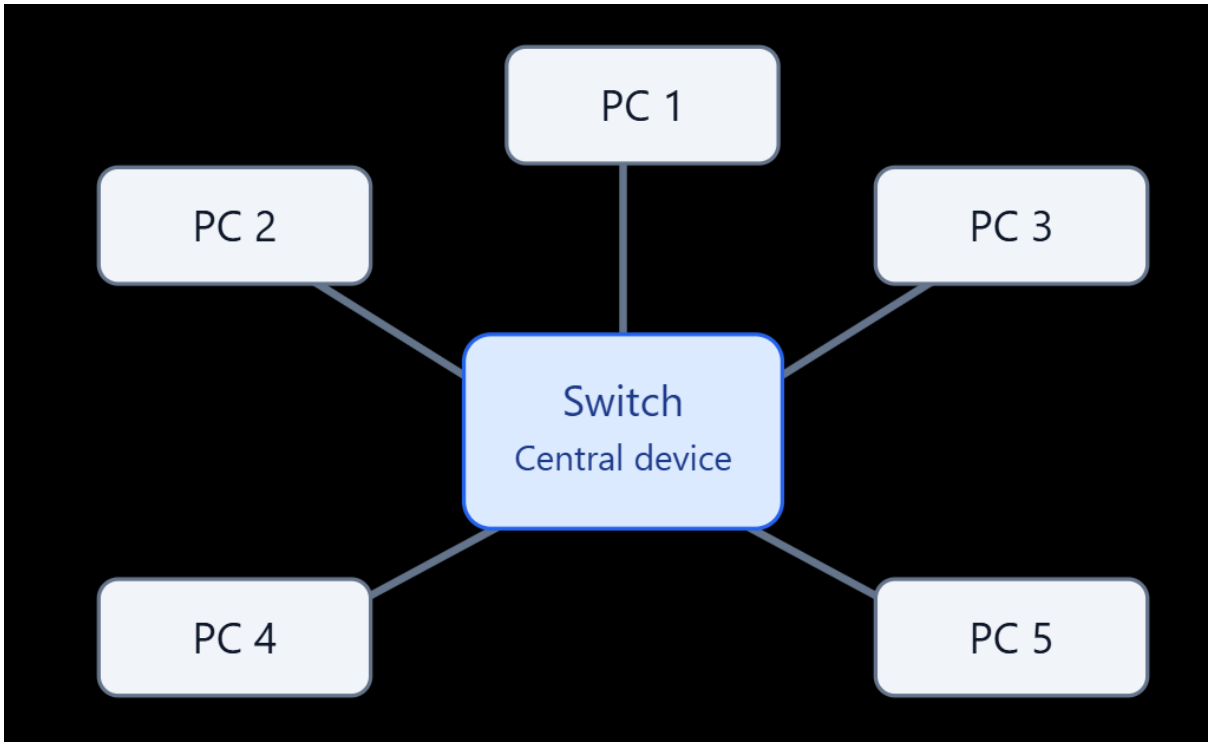
Network Topologies

1. Star, Bus, and Mesh Topology Diagrams

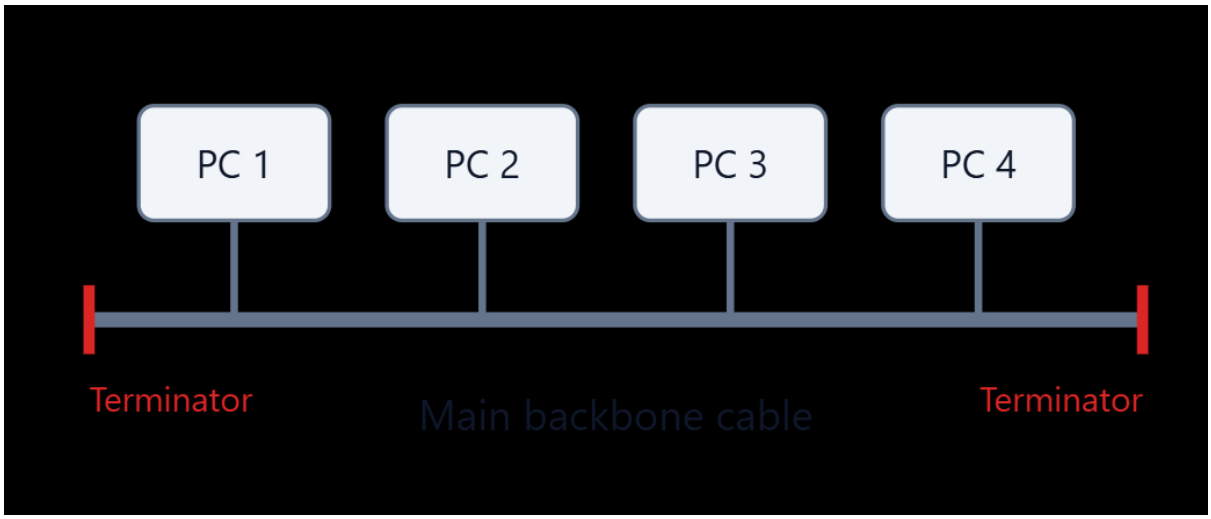
A. Star Topology



Each computer connects to one central switch.

Example: Flipkart's office network could use a star topology to connect employee computers to a central switch. It is easy to manage and troubleshoot.

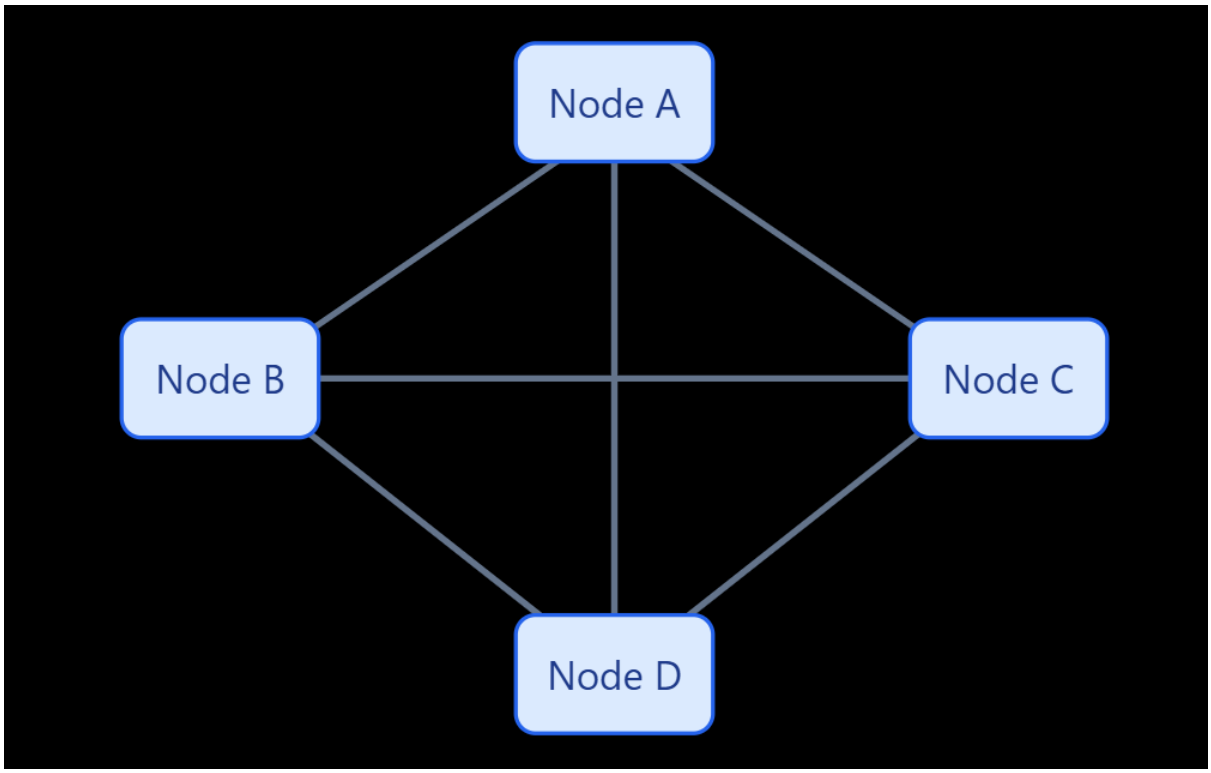
B. Bus Topology



All computers share one main cable.

Example: A small, older computer lab could use a bus topology because it requires less cable. However, a backbone cable failure can disrupt the entire network.

C. Mesh Topology



Nodes have multiple direct connections to one another.

Example: A distributed server network supporting WhatsApp could use mesh-like connections between data centers. Multiple paths help maintain connectivity if one link fails.

2. Comparison of Network Topologies

Topology	Advantages (4)	Disadvantages (4)	Best Use Case
Star	1. Easy to install. 2. Easy to troubleshoot. 3. One computer failure doesn't affect others. 4. Easy to expand.	1. Requires more cables. 2. Needs a central switch or hub. 3. Central device failure affects the whole network. 4. Installation costs can be higher.	Office networks using computers for Flipkart order processing.
Bus	1. Uses less cable. 2. Low installation cost for small setups. 3. Simple layout.	1. Backbone failure can stop the network. 2. Performance decreases with heavy traffic.	Small, older wired networks; not ideal for modern Instagram-scale services.

	4. No central switch required.	3. Troubleshooting can be difficult. 4. Limited scalability.	
Ring	1. Organized data flow. 2. Can reduce data collisions in traditional token-ring networks. 3. Predictable access in token-based systems. 4. Can provide predictable performance under controlled traffic.	1. A single link or node failure may disrupt a basic ring. 2. Adding devices can interrupt operation. 3. Troubleshooting may be difficult. 4. Expansion is less flexible.	Specialized ring-based industrial networks or network infrastructure.
Mesh	1. Highly reliable. 2. Multiple paths for data. 3. No single link must carry all traffic. 4. Supports fault tolerance.	1. Expensive to install. 2. Requires many connections and cables. 3. Configuration is complex. 4. Maintenance can be difficult.	Interconnected data centers and network infrastructure supporting services such as Paytm.

Note: Instagram, Zomato and Paytm are applications, not individual network topologies. Their actual infrastructure can combine several topologies and routing designs.

3. Best Topology for a Multiplayer Mobile Game

Chosen topology: Star topology (client-server architecture).

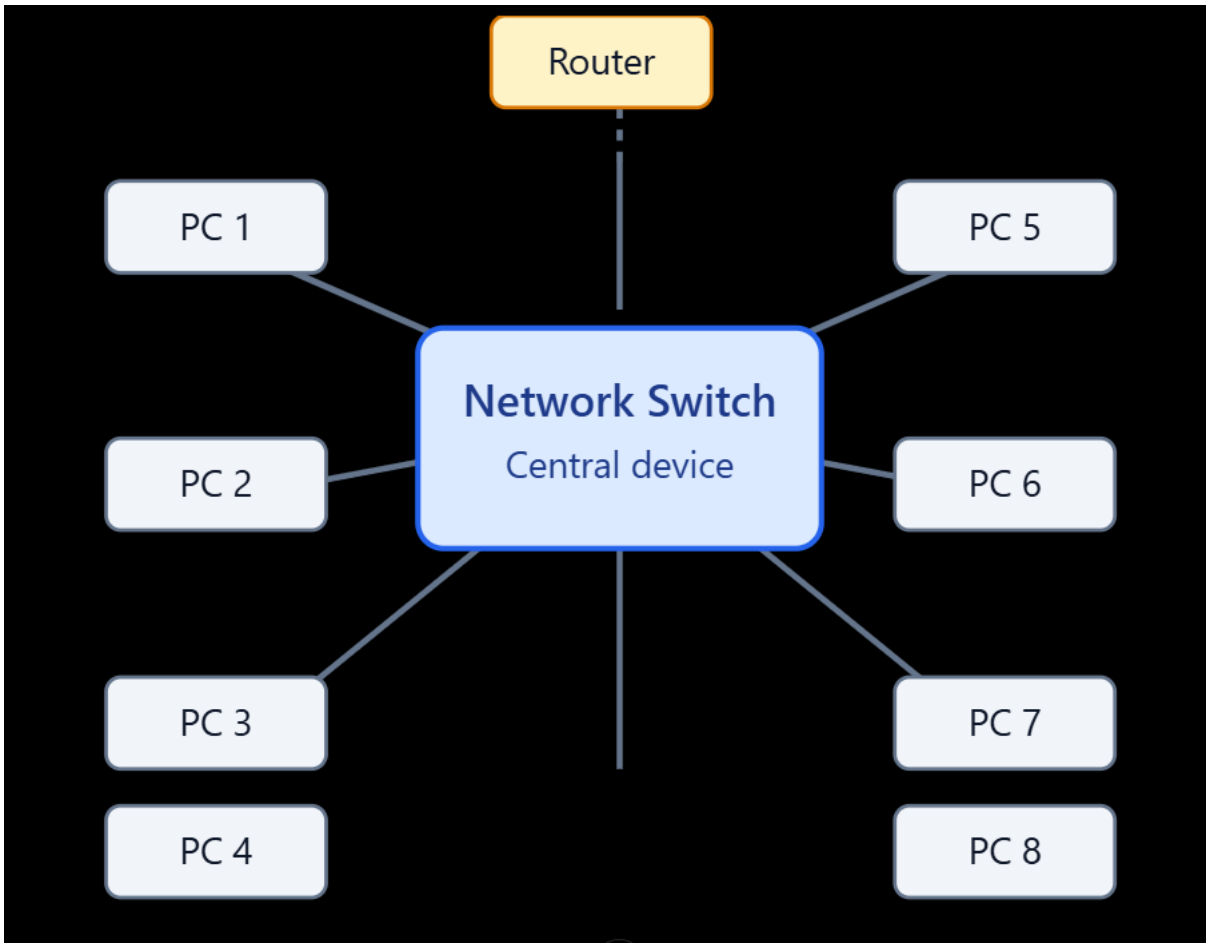
1. All players' devices connect to a dedicated game server, which manages player actions and game state.
2. A central server can process movements and updates efficiently, helping reduce delays when it is located close to players.
3. If one player's device disconnects, the other players can continue playing; however, a server failure can affect the whole match.
4. The setup is generally more affordable than a fully connected mesh between every player, although reliable servers and backup systems add cost.

Example: A PUBG-style game can use dedicated regional servers to improve responsiveness for nearby players.

4. Network Topology for a Small Warehouse

Recommended topology: Star topology

A small Flipkart-style warehouse needs to connect 8 computers for order processing and inventory updates.



Eight computers connect to a central switch; a router can provide Internet access.

Reasoning:

- Easy troubleshooting: If one computer or cable fails, the others continue working.
- Easy expansion: More computers can be added by connecting them to the switch, provided enough ports are available.
- Cost-effective: A single switch and Ethernet cables provide a practical, reliable network for a small warehouse.